

HEALTH CARE

Health care courses offer students an opportunity to investigate factors contributing to personal health and, at the same time, gain an introduction to a range of careers in the health care industry. These careers include, but are not limited to, child care worker, dental assistant/dentist, gerontologist, laboratory technician/technologist, doctor, nursing assistant/nurse, pharmacy assistant/pharmacist, and personal support worker. Courses in health care help prepare students for rewarding careers in a sector that is set to expand rapidly as our population ages.

Students will gain hands-on experience using industry-standard instruments, equipment, and materials and practising current techniques. They will learn the professional terminology of the field, acquire transferable problem-solving skills, and expand their communication and interpersonal skills as they interact with their peers and clients in a variety of simulated care scenarios. As they acquire theoretical understanding and practical skills in their areas of interest, students will become better prepared to make informed career choices.

The list of approved emphasis areas for health care can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Health Care, Grade 12

University/College Preparation

TPJ4M

This course focuses on the development of a range of skills needed to analyse and interpret clinical findings. Students will learn about accepted health care practices and demonstrate an understanding of basic procedures and the use of appropriate instruments and equipment. They will acquire an understanding of basic concepts related to the function of the human immune system and explore the relationship between pathology and disease prevention and treatment. Students will expand their awareness of workers' health and safety issues, analyse environmental and societal issues related to health care, and further explore professional practice standards and postsecondary destinations in the field.

Prerequisite: Health Care, Grade 11, University/College Preparation

A. HEALTH CARE FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** demonstrate an understanding of health care terminology and its correct usage;
- A2.** demonstrate an understanding of the structure and functioning of the immune system;
- A3.** demonstrate an understanding of fundamental concepts in pathology and their application to the diagnosis and treatment of disease;
- A4.** demonstrate an understanding of the transmission of disease and methods of preventing it.

SPECIFIC EXPECTATIONS

A1. Health Care Terminology

By the end of this course, students will:

- A1.1** demonstrate an understanding of correct terminology for equipment, instruments, and procedures (*e.g., vacutainer blood tube, electrocardiogram, culture tubes/slides, tri-flo syringe, otoscope, magnetic resonance imaging [MRI]*), and use it correctly in oral and written communication;
- A1.2** demonstrate an understanding of medical terminology used in vital sign analysis (*e.g., febrile, auscultate, antecubital space, tachypnea, inflammation*), and use it correctly in oral and written communication;
- A1.3** identify and use medical abbreviations correctly (*e.g., MI [myocardial infarction], bid [two times per day/bis in die]*).

A2. The Immune System

By the end of this course, students will:

- A2.1** demonstrate a basic understanding of the anatomy and physiology of the immune system (*e.g., location and function of bone marrow, thymus gland, lymph nodes, spleen; role of white blood cells and antibodies in response to pathogens and antigens*);
- A2.2** describe in detail the signs and symptoms of inflammation;
- A2.3** describe the signs and symptoms of common immune disorders (*e.g., lupus, rheumatoid arthritis, chronic fatigue syndrome*), and identify through research possible treatment strategies.

A3. Pathology

By the end of this course, students will:

- A3.1** explain how microorganisms cause disease, and identify factors that affect the development of a disease (*e.g., bacterial growth requirements, virulence*);
- A3.2** analyse the life cycle of microorganisms and their impact on health care;
- A3.3** explain how microorganisms are classified (*e.g., as bacteria, fungi, viruses, parasites*);
- A3.4** describe causes of various infectious diseases (*e.g., middle ear infections are caused by bacteria, viruses, or allergies*), and describe the symptoms related to each (*e.g., elevated temperature, inflammation, pain*);
- A3.5** analyse how pathogens can affect multiple body systems (*e.g., streptococcus causes strep throat and can move to the heart, kidneys, and joints; ruptured appendicitis can cause peritonitis, septicemia, and death; dental bacteria can cause cavities and heart disease*);
- A3.6** describe abnormal values for each type of vital sign, and identify possible causes and appropriate interventions for each of these values (*e.g., elevated temperature can be caused by gram-positive cocci and may require antibiotic treatment*);
- A3.7** describe common diagnostic methods for identifying diseases (*e.g., urinalysis, complete blood count, X-rays, scans*).

A4. Disease Prevention and Treatment

By the end of this course, students will:

- A4.1** demonstrate an understanding of disease transmission (*e.g., the chain of infection*) and the role of the health care provider in health promotion and in controlling the spread of infection (*e.g., taking standard infection control precautions, such as following hand hygiene rules, sterilizing instruments and equipment, and keeping client areas clean*);
- A4.2** identify and describe characteristics of various disease outbreaks (*e.g., outbreaks of methicillin-resistant staphylococcus aureus [MRSA], C. difficile, severe acute respiratory syndrome [SARS], avian flu*);
- A4.3** identify methods of dealing with the outbreak of a disease using the chain of infection (*e.g., isolation protocols*);
- A4.4** describe common procedures and strategies for the prevention or early detection of disease (*e.g., good dental hygiene, health screening procedures such as Pap smears, examinations such as self-breast examination (SBE), immunizations such as chicken pox and hepatitis A/B vaccinations*);
- A4.5** demonstrate an understanding of Rh and ABO blood types and describe how incompatibility issues in blood transfusion (*e.g., fetal Rh incompatibility*) can be avoided;
- A4.6** identify examples of the misuse of antibiotics and antibacterial substances (*e.g., in household products*) and describe the consequences (*e.g., antibiotic resistance, superbugs*).

B. HEALTH CARE SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** demonstrate competence in using health care instruments, equipment, and materials;
- B2.** demonstrate the ability to use vital signs to determine a client's health status;
- B3.** demonstrate an understanding of and apply standard practices and procedures used in the health care field;
- B4.** demonstrate an understanding of and apply communication and documentation techniques required in the health care industry.

SPECIFIC EXPECTATIONS

B1. Instruments, Equipment, and Materials

By the end of this course, students will:

- B1.1** identify a wide range of tools, equipment, and materials used in the health care industry (*e.g., forceps, sterile dressing trays/packages, centrifuges, autoclaves, reagent strips, glucometers, containers for culture and sensitivity testing*), and use them safely and correctly;
- B1.2** prepare, clean, and store instruments, equipment, and materials, following standard industry procedures (*e.g., follow standard medical and surgical aseptic procedures while handling equipment such as stethoscopes, thermometers, and scalers*).

B2. Vital Signs

By the end of this course, students will:

- B2.1** demonstrate the ability to use equipment to measure the four primary vital signs (temperature, pulse, respiration, blood pressure) and to assess the degree of pain (often considered an additional vital sign);
- B2.2** assess and document a client's vital signs: temperature (by oral, axillary, and tympanic routes), pulse rate (rhythm and volume), respiration rate (rhythm and character), blood pressure, and pain intensity (*e.g., using a pain scale*);
- B2.3** identify normal values for temperature, pulse, respiration, blood pressure, and pain intensity

across the lifespan, and assess possible health implications of abnormal values (*e.g., elevated temperature suggests a possible infection; an elevated blood pressure reading may be an early indication of hypertension*).

B3. Practices and Procedures

By the end of this course, students will:

- B3.1** demonstrate an understanding of (*e.g., through role play*) and apply standard first-aid (SFA) and cardiopulmonary resuscitation (CPR) procedures (*e.g., emergency scene management; control of bleeding; treatment of sprains, fractures, burns, loss of consciousness, anaphylactic reaction/shock, choking*);
- B3.2** use appropriate medical aseptic procedures (*e.g., hand hygiene, gloving, proper use of biohazard waste containers and handling of contaminated laundry waste, steam sterilization of equipment*) to prevent the spread of pathogens;
- B3.3** use appropriate precautions when handling body substances, as demonstrated through role play and the use of simulated body products (*e.g., blood products, body fluids, human tissue, and materials contaminated with these substances*);
- B3.4** demonstrate an understanding of and apply standard industry procedures for handling specimens and administering medications and treatments (*e.g., safely collect, label, store, and transport specimens to prevent contamination and degradation of sample; correctly dispense mock medication following the College of Nurses of*

Ontario [CNO] rights of medication administration standards [right time, right client, right medication, right reason, right dose, right frequency, right route, right site, right documentation]; administer heat and cold applications as appropriate);

- B3.5** analyse and interpret findings from simulated client information (*e.g., use case studies to complete a history, identify symptoms, determine a diagnosis, and prepare a treatment plan*).

B4. Communication and Documentation

By the end of this course, students will:

- B4.1** demonstrate an understanding of and apply effective therapeutic communication techniques (*e.g., active listening, paraphrasing, use of culturally responsive communication techniques*) in a variety of situations;

- B4.2** collect pertinent information in a simulated patient care situation and report patient care data on the appropriate forms (*e.g., flow charts, graphic charts such as vital signs records, dental charts*);

- B4.3** assess communication challenges related to cultural and religious diversity (*e.g., the need to respect religious beliefs such as those that preclude blood transfusions and after-death care, and cultural beliefs related to child-birth and pain management or to the care of female patients by male doctors or nurses*), and apply strategies for addressing them.

C. HEALTH CARE, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** assess the impact of the health care industry on the environment, and identify legal requirements and guidelines for protecting the environment from harmful consequences;
- C2.** assess the societal and ethical implications of advances in medical technology.

SPECIFIC EXPECTATIONS

C1. Health Care and the Environment

By the end of this course, students will:

- C1.1** describe the environmental impact of technological advances in the health care field (*e.g., single-use devices create end disposal issues, as incineration results in heavy metal toxicity and landfilling creates biohazardous leachate; because of the increased use of pharmaceuticals, traces of prescription drugs are now being found in surface water*);
- C1.2** research and report on laws, regulations, guidelines, and information sources pertaining to the disposal of medical waste (*e.g., Atomic Energy Control Board – radioactive materials; Workplace Hazardous Materials Information System [WHMIS] – chemical hazards*);
- C1.3** identify some environmental impacts of a health care product over its complete life cycle (*e.g., use of harmful chemicals in manufacturing the product, energy consumption for manufacturing and transportation of raw materials and completed products, packaging waste, impact of the product when disposed of, impact of drug residues in human body wastes*).

C2. Health Care and Society

By the end of this course, students will:

- C2.1** analyse ethical issues related to various products and procedures that have contributed to improved health care (*e.g., life-extending devices such as cardiac pacemakers and internal defibrillators, microscopic/robotic surgical procedures; priorities for organ transplants*);
- C2.2** assess the impact of medical biotechnology (*e.g., in vitro fertilization [IVF], chorionic villus sampling [CVS], umbilical cord blood storage*) on people's everyday lives;
- C2.3** identify and report on bioethical issues in health care (*e.g., right to refuse treatment, euthanasia, stem cell research*).

D. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** demonstrate an understanding of and comply with safe working practices and the laws and regulations governing the health and safety of workers in the health care industry;
- D2.** demonstrate an understanding of legal, ethical, and professional standards governing the practice of health care;
- D3.** demonstrate an understanding of the range of career opportunities in conventional and alternative therapy fields.

SPECIFIC EXPECTATIONS

D1. Health and Safety

By the end of this course, students will:

- D1.1** explain the principal components of and comply with legislation, regulations, and guidelines pertaining to the safety of the health care workplace (e.g., *Occupational Health and Safety Act [OHSA]*, *Workplace Hazardous Materials Information System [WHMIS]*);
- D1.2** identify potential hazards in the health care workplace (e.g., *biological, chemical, radiological, and physical hazards such as body fluids, soiled linens, cleaning agents, X-rays, combative clients, and sharp objects*);
- D1.3** assess various workplace factors that may lead to short-term or long-term health and safety issues for health care workers (e.g., *long hours on feet, long work shifts, lifting/moving of patients, ergonomic considerations*);
- D1.4** demonstrate an understanding of and apply safe procedures for the use of instruments, materials, and equipment as outlined in safety manuals, operating instructions, and institutional requirements;
- D1.5** use protective clothing and equipment as required to keep themselves and others safe and free from harm.

D2. Professional Standards

By the end of this course, students will:

- D2.1** explain the principal components of legislation regulating the practice of health care in Ontario (e.g., *Regulated Health Professions Act [RHPA]* sets the framework for regulating the scope of practice of health professions);
- D2.2** assess malpractice and other liability issues (e.g., *termination of care, negligence, misuse of controlled substances, misuse of power of attorney*);
- D2.3** identify the governing bodies for various health care professions (e.g., *College of Nurses of Ontario, College of Physicians and Surgeons, College of Dentistry*) and explain their roles and responsibilities;
- D2.4** identify the steps involved in reporting professional misconduct to a professional governing body or college (e.g., *College of Physicians and Surgeons, College of Nurses of Ontario, College of Dentistry*) and describe possible outcomes (e.g., *suspension, loss of licence, retraining*);
- D2.5** demonstrate an understanding of the importance of maintaining client confidentiality, and explain the obligations of health care providers under the Personal Health Information Act for ensuring client privacy and confidentiality (e.g., *requirements regarding the collection, use, retention, disclosure, and disposal of client information, such as diagnostic test results or HIV status*);

D2.6 demonstrate an understanding of how medical knowledge is advanced (*e.g., biological research, epidemiological research, drug research and trials, accumulated experience*), and identify ways in which individuals can keep abreast of advancing medical knowledge (*e.g., reading journal articles, attending seminars and workshops, receiving in-service training*).

D3. Career Opportunities

By the end of this course, students will:

D3.1 assess health care career opportunities in conventional and alternative therapy fields and describe the education and/or training required for a specific career path;

D3.2 demonstrate an understanding of and apply the Essential Skills that are important for success in the health care industry, as identified in the Ontario Skills Passport (*e.g., computer use, data analysis, job task planning and organizing*);

D3.3 demonstrate an understanding of and apply the work habits that are important for success in the health care industry, as identified in the Ontario Skills Passport (*e.g., teamwork, organization, self-advocacy*);

D3.4 maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in health care (*e.g., Passport to Safety certificate, reports, assignments, Ontario Skills Passport Work Plan and Transition Plan*), and explain why having a current portfolio is important for career development and advancement.